



Hope Kim @rlaope

Software Engineer

In 2014, a kid who dreamed of being a scientist in a white lab coat discovered computers. I don't believe in limits that haven't been mathematically proven. The harder the problem, the more exciting.

At 17, I took on a government-funded project—unfamiliar domains, tight deadlines and resource constraints, everyone thought it was impossible. Under pressure, I dove deeper into system development, grew alongside my teammates, secured \$33K investment, and built a new project room at Software Meister High School.

HISTORY

2014. Junior

First software development in elementary school. Built 'FLY', an RPG game.

Software Meister High School (CSE). 2022 ~ 2024

Researched and validated server systems, JVM internals, and open-source systems.

10+ Projects.

School OAuth, nationwide dormitory management, hackathon AWS award, SW-Wave booth.

Government Project. 2023. 09 ~ 2024. 11

Vocational education platform for specialized high schools. Secured 45M KRW investment.

HonestAI. 2024. Dec ~ Now

Server Developer at Vertical AI fintech. B2B BaaS, JVM/K8s observability, TCP tuning, Netty optimization, load testing.

OPEN SOURCE

Argus JFR-streaming profiler for JVM 21+. Virtual Threads monitoring, lock-free ring buffer, WebSocket dashboard.	kar98k Go traffic simulator for chaos engineering. HTTP/1.1, HTTP/2, gRPC. Poisson spikes, TUI, P50/P95/P99.	claude-code-savant Claude Code plugin. 5 personas, 16 dev agents, 14 biz agents. Web chat UI.
airML Lightweight ML runtime for ONNX. ~50MB, 0.01s cold start. Apple Silicon native.	Mascot IntelliJ IDEA mascot plugin. Published on JetBrains Marketplace.	cache-labs Cache design study. Local + Redis dual-layer, Circuit Breaker, high-traffic.
g1-log-capture G1GC log analyzer. STW pause detection, tuning suggestions.	tcp-css-loadtest TCP mock server for financial load testing. Nice protocol, EUC-KR.	TatVTON Diffusion tattoo fitting. SAM 2 + SDXL + ControlNet + IP-Adapter. 12GB VRAM.

PRESENTATION

KotlinConf'24 Global. 2024

JetBrains KotlinConf Global in South Korea. Project Valhalla, JVM Type System, Value Class Benchmark.

IT Union Conference. 2025

Meister Developer IT Union. Carpe Diem – Software development skills and individuality.

GSM DevConf. 2023

Growth of Developer – Developer's growth curve, direction, and the mindset to pursue.

INTERESTED

JVM	GC	SRE	AI (ML/DL)	Cloud	Network	OS	Art	The act of satisfying someone through engineering
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CONTACT

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HonestAI

Software Engineer – develop product, server/infra engineering 2024. Dec ~ 2026. Mar

A detailed document describing the problems, considerations, solutions, and results encountered while working at HonestAI. Responsibilities and experiences are in the Resume below; detailed descriptions are narrowed down to three topics.

Resume

SERVER

- Developed/operated savings bank online linked investment system
- Integrated external APIs: Savings Bank Central, NICE/KCB, Shinhan Bank, KFTC, Toss, KakaoPay
- Loan system traffic **1,000 → 500K** load testing & infra scaling, resolved Netty server bottleneck (jstack, netty)
- Built linked investment fault alert system, **fault detection rate improved (40% → 90%)**
- Built fault response process for investment inquiry → confirmation → deposit → execution cycle (**Read-Time 15m → 30s**)
- NICE credit evaluation batch query data **100MB → 1GB** expansion: bandwidth saturation response & traffic control
- TCP tuning on financial messaging server, latency improvement (connection reuse socket pool, Nagle algorithm disable, sub-ms level)

INFRA

- Built JVM, K8s Monitoring & Observability with Prometheus, Grafana, Micrometer
- Built log Observability with Loki, Grafana dashboards
- Athena log table redesign with Partition Projection, **query time improved (4~5m → 1~4s)**
- Legacy JVM Observability setup (jmx-prometheus) & tuning (jcmd, jstack, jmap etc.)
- Detected settlement batch **Full GC Hang**, memory adjustment & Hang detection alert (alert manager) system
- Platform server G1GC migration (**P99 2~3s → 0.4s**) & OpenJDK 8 optimization verification, comparative benchmarks
- Gradle build parallelism/cache & daemon heap tuning, CI flow improvement (**build time 20m → 1~3m**)
- Micrometer-based domain error aggregation for fault frequency & priority analysis (also saved Slack API tokens)
- Core banking task DB lock timewait, deadlock monitoring & system improvement

FINANCE

- Supported syncing HonestFund core banking bond ledger with savings bank linked investment bond ledger
- Managed regular repayment tasks, special realization & acceleration-related bond schedule operations
- Managed credit bureau, NICE/KCB, KFTC bond status sync & concurrent loan registration system
- Managed loan application-to-disbursement process & **developed auto-loan system based on R-Clips credit scoring model**

For detailed descriptions → rlaope.github.io/honest.html

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Last updated 2026. 3. 11



Government Project

Backend Lead – Vocational Education Platform for Specialized High Schools 2023. 09 ~ 2024. 11

At 17, I jumped into a government-funded project. Built a vocational education & credit management platform (School-of-Company) for specialized high school students.

Led server development and collaborated with stakeholders from affiliated organizations.

Resume

SERVER

- Developed/led vocational education & credit management server system
- JPA Persistable, hashCode(), equals() override, **N+1 query optimization** (prevented extra queries on create)
- Led server development and technology decision consensus from initial system design to production
- Wrote Architecture Decision Records, facilitated technical decision consensus with stakeholders
- **80+ partner companies, 16+ affiliated agencies** – developed new features reflecting VoC periodically

QA

- Conducted end-to-end integration process testing from signup to course enrollment to credit management
- QA data set time (15 times/day, ~5min each) automation → **improved to ~5 second simulation environment**

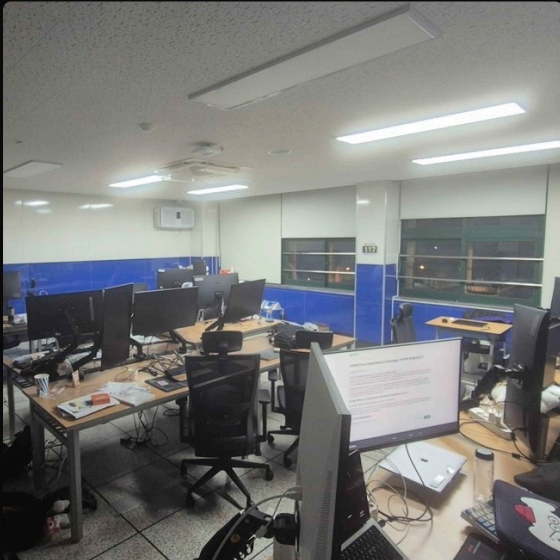
INFRA & PRODUCTIVITY

- Assigned code review PR priority, resolved PR neglect with Discord Bot-based Reminder
- Designed entire server infra, **Terraform infrastructure-as-code**, handled deployment/operations then handed off to team

Achievements

- + Government-funded project – Board of Education specialized high school credit certification platform
- + After successful project delivery, **secured \$33K (45M KRW) investment**
- + Built a new project room at Software Meister High School
- + Handed off operations to the next cohort

School Company Room



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